

Colonization of butchers with livestock-associated methicillin-resistant Staphylococcus aureus.

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Reports have documented colonization of swine in Europe, North America and more recently in China with livestock-associated methicillin-resistant *Staphylococcus aureus* (LA-MRSA). Contamination of pig farmers, veterinarians and abattoir workers with these strains has been observed. However, although contamination levels of 10% of retail pork were reported from the Netherlands and Canada, there are limited data of contamination rates of workers handling raw meat. We investigated the rates of MRSA contamination of local butchers working in wet markets, where recently slaughtered pigs are cut up. Nasal swabs collected from 300 pork butchers at markets throughout Hong Kong were enriched in brain heart infusion broth with 5% salt and cultured on MRSASelect(®) . Isolates were confirmed as *Staphylococcus aureus* and susceptibility testing performed. The presence of *mecA* was confirmed, SCCmec and *spa* type determined and relatedness investigated by PFGE. Subjects completed a questionnaire on MRSA carriage risk factors. Seventeen samples (5.6%) yielded MRSA, 15 harbouring SCCmec IVb. Ten strains were t899 (CC9), previously reported from local pig carcasses. Five strains were healthcare associated: SCCmec type II, t701(CC6), colonizing two subjects at the same establishment, and single isolates of t008 (CC8), t002 (CC5) and t123 (CC45). The remaining isolates were t359 (CC97), previously reported from buffaloes, and t375 (CC5), reported from bovine milk. None of these butchers reported recent hospitalization or a healthcare worker in the family. Two had recently received antibiotics, one for a skin infection. Four reported wound infections within the last year. All were exposed to meat for >9 h per day. Carriage of MRSA was higher in butchers than in the general community. Although five strains were probably of healthcare origin, the high incidence of t899 (CC9) suggests that cross-contamination from pork occurs frequently. Washing of hands after touching raw pork is advised. © 2012 Blackwell Verlag GmbH.